

GRB 090620

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ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 090620 (*Fermi*/GBM trigger bn090620400), the fifth burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline. The burst is compact ($T_{90} = 12.50^{+0.42}_{-0.57}$ s) and is the first of the campaign in which all three minimum-variability estimators return significant measurements (burst #1’s windowed value carried a $z = 2.5$ caveat): the canonical windowed method finds 28.2 ± 2.3 ms, the Haar cross-check 353 ± 98 ms, and the global wavelet method 215 ± 17 ms — the *same* grid-quantized value the wavelet estimator returned for three earlier campaign bursts (#1, #2, and the unpapered #3), hardening the campaign’s finding that cross-burst CWT values cluster on the discrete wavelet-scale ladder. The pulse breaks the campaign’s FRED monotony: the two-sigmoid asymmetry is $\varphi = 0.40 \pm 0.05$ ($R^2 = 0.94$), classified mixed rather than FRED-like. The window-scanned spectral lag is $\tau = 0.68^{+0.05}_{-0.06} \pm 0.13_{\text{win}}$ s (25–50 vs. 100–300 keV, soft lagging hard). Spectroscopy over nine Bayesian blocks plus the integrated spectrum repeats the epoch pattern of burst #4: the first four blocks sit at or above the synchrotron line of death (Band $\alpha = +0.43$ to $+0.10$) while the decay falls away from it ($\alpha = -0.87$ by the last block), and E_p tracks the light curve non-monotonically (196 $\rightarrow$ 237 $\rightarrow$ 56 keV). The integrated spectrum is won by a single-break SBPL ($\alpha = -0.46$, $E_b = 131$ keV, $\beta = -2.98$; Band within $\Delta\text{AIC} = 1.3$). Blackbody-bearing models win three decay blocks, but the campaign’s edge gate disarms the sharpest of them: the block-7 blackbody sits at $kT = 1.10 \pm 0.09$ keV, whose Wien peak (4.3 keV) falls below the 8 keV fit boundary — the defining signature of the edge-artifact class, reported as such and not as a detection. The 24×10 model grid renders under the no-model-dropped rule with two engine-refusal cells labeled (Band+PL, integrated; BandR+CPL, block 1); one effective-area factor rails (n_b at 0.80) and is disclosed.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 090620 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2009 June 20. As burst #5 of the single-pulse campaign it runs on the pipeline frozen from bursts #1–#2. Its role in the accumulating cross-burst picture is threefold: it is the first burst where the three variability estimators all return significant values (giving the estimator-dependence question its first fully significant row); it is the first non-FRED pulse shape; and it repeats — in nine finer blocks — the epoch-level synchrotron-violation pattern that burst #4 first showed, strengthening the case that the line of death is crossed *within* bursts, not between them.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_6 , n_7 , n_8 , n_b and BGO b_1 (Figure 1), with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the

30–40 keV K-edge, BGO 0.2–38 MeV. Background and source selections are the human-approved Stage-1 decisions (source window $(-0.91, 18.27)$ s; Figures 2–3). Effective-area factors float in $[0.8, 1.2]$; in the integrated winner n_b rails at 0.80 while the others float ($n_7 = 0.86$, $n_8 = 0.87$, $b_1 = 0.92$) — a single-rail configuration, milder than burst #4’s, disclosed as a standing caveat.

3. TIMING ANALYSIS

All timing runs on the approved selections, reference NaI n_7 .

3.1. Durations

The windowed duration is $T_{90} = 12.50^{+0.42}_{-0.57}$ s spanning $[0.88, 13.37]$ s (covariance-true Monte Carlo; window not truncated), with $T_{50} = 4.15$ s. $T_{50}/T_{90} = 0.33$: a far more symmetric energy release than burst #4’s 0.51, anticipating the pulse-shape result.

3.2. Pulse Shape

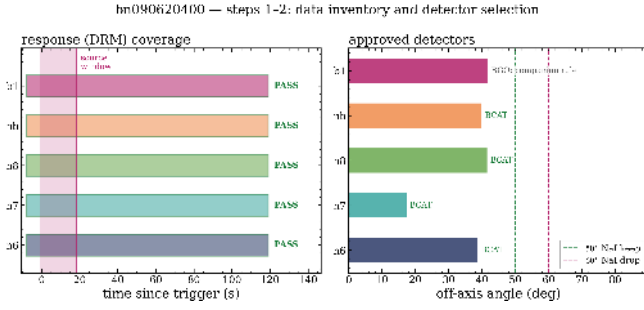


Figure 1. Detector selection for GRB 090620: source angles with the 50° rule; selected detectors n_6 , n_7 , n_8 , n_b , b_1 .

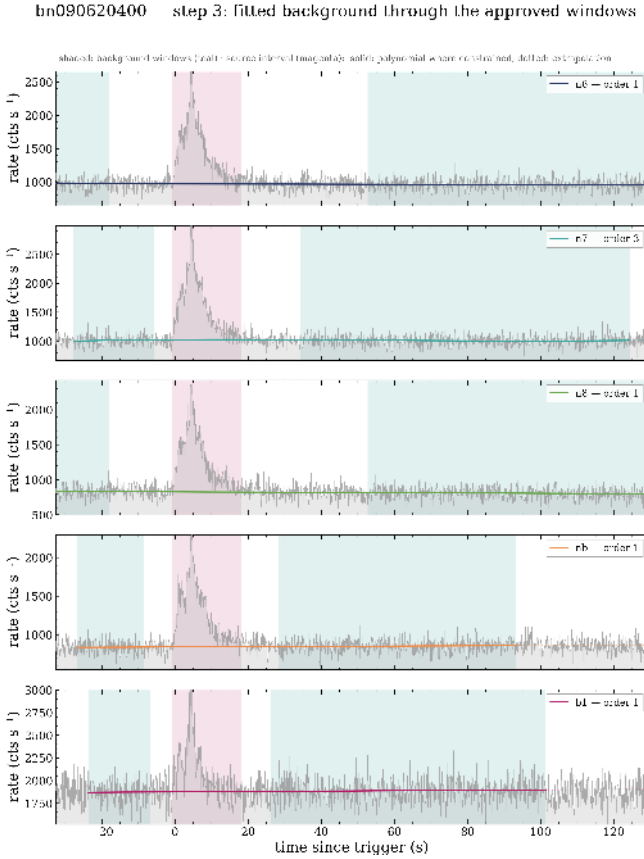


Figure 2. Approved background intervals and per-detector polynomial fits.

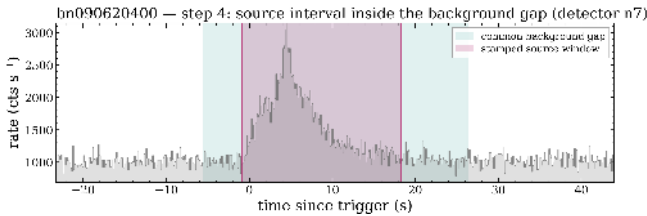


Figure 3. Approved source interval over the background-subtracted light curve.

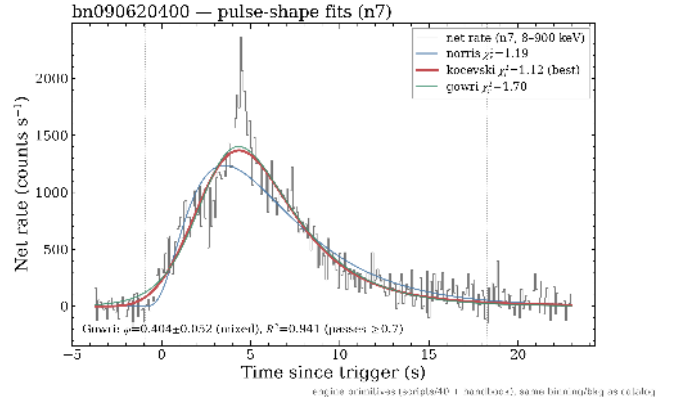


Figure 4. Pulse-shape fits to the net light curve; Kocevski wins and the two-sigmoid asymmetry $\varphi = 0.40 \pm 0.05$ is the campaign's first mixed (non-FRED) classification.

The Kocevski et al. (2003) profile wins the shape comparison, and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.40 \pm 0.05$ ($R^2 = 0.94$) — classified *mixed*, not FRED-like: the first departure from the FRED family in the campaign (bursts #1, #2, #4: $\varphi = 0.317, 0.263, 0.017$). The sample's pulse-shape axis now spans a factor of 24 in φ (Figure 4).

3.3. Minimum Variability Timescale

For the first time in the campaign all three estimators return significant measurements (Figure 5; burst #1's windowed value was quoted with a $z = 2.5$ caveat), and they disagree by an order of magnitude in a structured way: the windowed framework of Bala et al. (2026) — the engine-selected canonical value — finds 28.2 ± 2.3 ms ($\Delta t = 2$ s segment, detection); the Haar cross-check (Golkhou & Butler 2014) finds 353 ± 98 ms; and the global CWT (Vianello et al. 2018) returns 215 ± 17 ms — the identical grid rung it returned for bursts #1, #2, and #3. Four bursts on one rung is no longer a coincidence: the CWT value is a property of the estimator's discrete scale ladder, not of the bursts, and the campaign quotes CWT only with this label. The estimator-independence assumption is falsified as stated in the campaign's assumption registry.

3.4. Spectral Lag

Scanning the cross-correlation fit window around the peak (half-widths 1.2–4.8 s; Figure 6) gives $\tau = 0.68^{+0.05}_{-0.06} \pm 0.13_{\text{win}}$ s (25–50 keV vs. 100–300 keV, soft lagging hard; detector n_7 , 16 ms resolution, peak significance 33σ); the scan members run 0.47–0.72 s and the half-spread is the quoted window systematic. The lag fraction $\tau/T_{90} \approx 5.4\%$ sits between burst #1's 8.5% and burst #4's 2.6%.

3.5. Per-band Durations

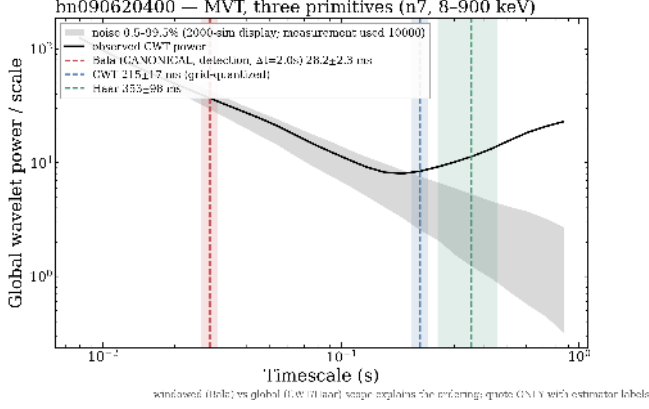


Figure 5. Minimum-variability estimators with explicit labels: canonical windowed 28.2 ± 2.3 ms; Haar 353 ± 98 ms; CWT on the recurrent 215 ms grid rung.

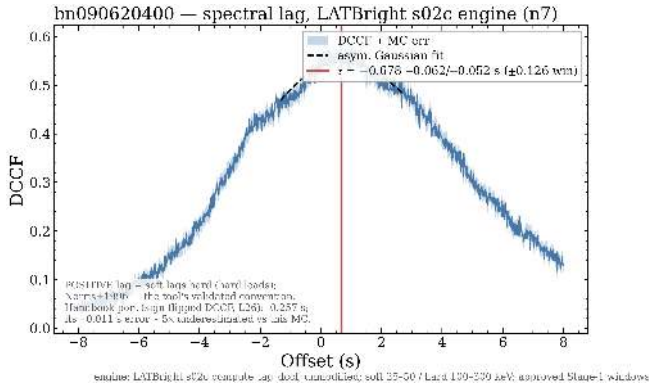


Figure 6. Cross-correlation lag with the window-scan systematic; the quoted value is the pulse-scaled member closest to the scan median.

Per-band light curves and the $T_{90}(E)$ relation are shown in Figure 7; the population slope of Qin et al. (2013) is drawn for reference only.

4. SPECTRAL BINNING

Nine Bayesian blocks plus the integrated spectrum (Figure 8) — the finest time resolution of the campaign so far — from the frozen 3ML Bayesian-block + significance hybrid (blocks computed 2026-08-12, reused unchanged).

5. TIME-RESOLVED AND TIME-INTEGRATED SPECTROSCOPY

5.1. Method

All 24 campaign models fit to every bin (PGstat; AIC ranking (Akaike 1974)); no LLE/LAT data exist for this burst (inclusion is gated on data availability, never detection significance). The full 24×10 grid renders under the no-model-dropped rule with two engine-refusal cells (Band+PL fails in the integrated spectrum;

BandR+CPL in block 1), shown as labeled refusal panels in the montages; per-bin all-model parameter tables accompany the products.

5.2. Integrated Spectrum

The integrated spectrum (0.05, 16.87) s is won by the single-break SBPL (Figure 9): $\alpha = -0.458^{+0.062}_{-0.056}$, $E_b = 131^{+18}_{-15}$ keV, $\beta = -2.98^{+0.17}_{-0.21}$ (AIC = 7277.6), with Band at $\Delta\text{AIC} = 1.3$ — a statistical tie between one-break shapes; unlike burst #4, the two-break DSBPL does not contend for the integrated spectrum here.

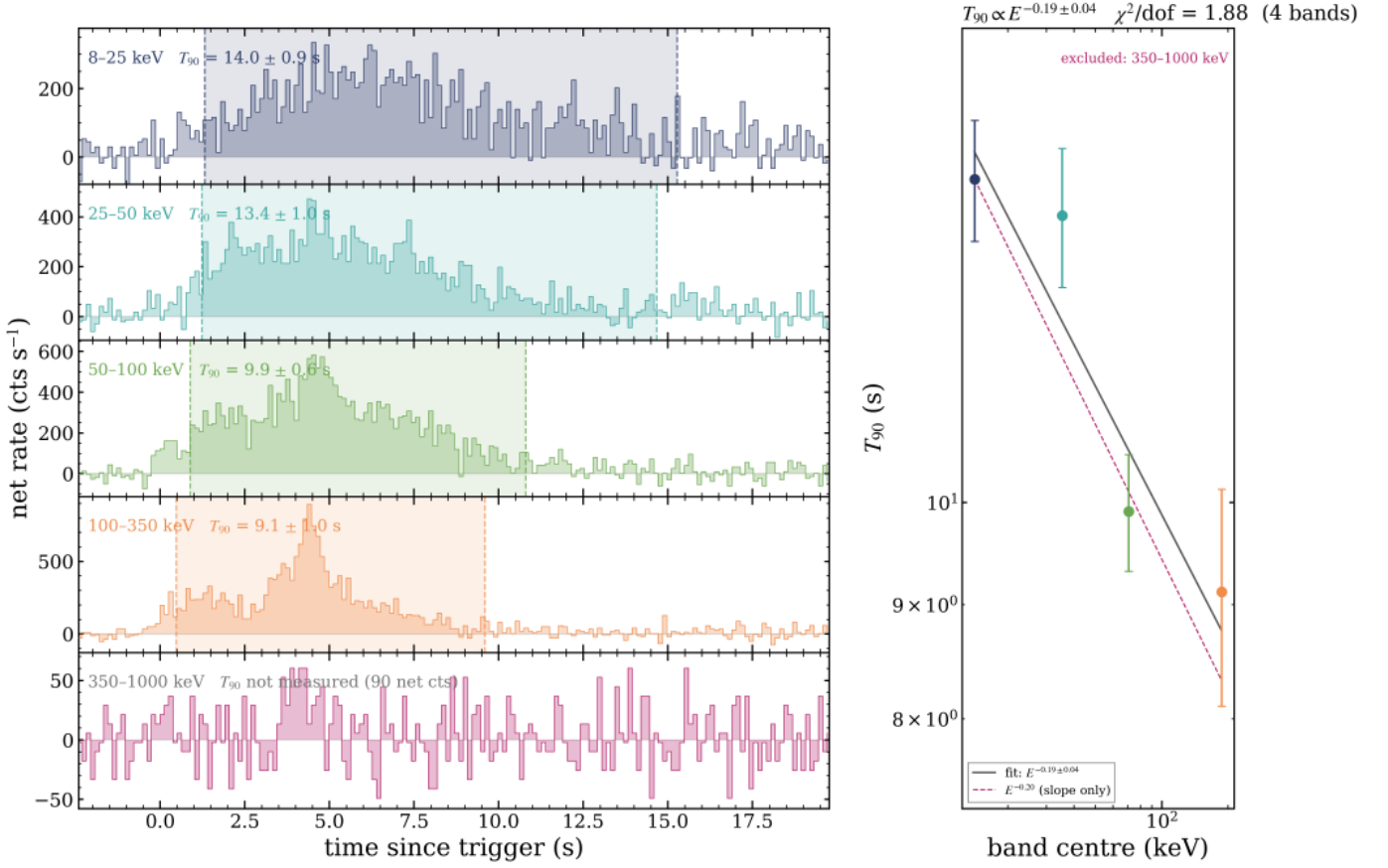
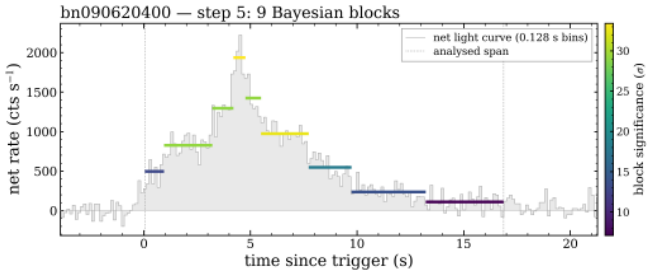
5.3. Time-Resolved Story

In the uniform Band convention α runs $+0.43, +0.28, -0.08, +0.10, -0.05, -0.13, -0.58, -0.05, -0.87$ across blocks 0–8 while E_p moves $196 \rightarrow 138 \rightarrow 215 \rightarrow 237 \rightarrow 181 \rightarrow 116 \rightarrow 100 \rightarrow 56 \rightarrow 56$ keV: the rise and peak blocks sit at or above the $-2/3$ line of death, the decay falls below it, and E_p tracks the light-curve structure rather than decaying monotonically. This repeats burst #4’s epoch pattern at finer time resolution (nine blocks against six): across the campaign the synchrotron violation is consistently an early-epoch property (Figure 10). Model preference is fragmented across the nine blocks (CPL, DSBPL, SBPL, Band, and BB-bearing winners each appear), with razor-thin margins in several blocks (bin 3: CPL over SBPL by $\Delta\text{AIC} = 0.2$) — at this time resolution the per-block statistics blur the model distinctions, and we lean on the block-to-block *trends* rather than any single-block winner.

5.4. Thermal Candidates and the Edge Gate

BB-bearing models win three decay blocks, and the campaign’s edge gate splits them decisively:

- Block 4: Band+BB, $kT = 12.4 \pm 1.1$ keV (valid; Wien peak 49 keV, edge-clean) — but the margin over plain Band is only $\Delta\text{AIC} = 0.4$, a statistical tie, so this is at most a weak candidate (Figure 11).
- Block 7: Band+BB, $kT = 1.10 \pm 0.09$ keV. The Wien peak $3.92 kT = 4.3$ keV lies *below* the 8 keV fit boundary: the blackbody is doing its work entirely where the data cannot see its peak, the defining signature of the edge-artifact class in the campaign’s lesson list. We report it as an edge artifact, not a thermal detection (Figure 13).
- Block 8: CPL+BB, $kT = 95 \pm 33$ keV (valid; margin over plain CPL $\Delta\text{AIC} = 1.1$, again a tie; Figure 12) — a high-side counterpart whose 35% temperature uncertainty carries little evidential weight.

bn090620400 — step 7: energy-resolved T_{90} (detector n7); dashed = t_5 , t_{95} **Figure 7.** Per-band light curves with individual T_{90} spans and the $T_{90}(E)$ relation.**Figure 8.** Bayesian-block spectral bins over the light curve.

Net: unlike burst #4's coherent cooling track, GRB 090620's blackbody preferences are block-local, marginal, and in the sharpest case an edge artifact. The campaign accumulator records zero robust thermal candidates for this burst.

6. QUALITY CONTROL AND PROVENANCE

Step-9 QC (Figure 14) is built from the canonical campaign fit table; every figure carries its producing script's SHA-256 in a JSON sidecar; numbers in this paper are recomputed from product files. All figures pass a fresh-context vision gate before delivery, enforced

by a pre-delivery hook. Fits come from the campaign's one-invocation 24-model engine run; the canonical table keeps the best optimizer minimum per (bin, model) across available invocations.

7. SUMMARY

1. GRB 090620 is a compact ($T_{90} = 12.50^{+0.42}_{-0.57}$ s) single pulse with the campaign's first *mixed* (non-FRED) shape classification ($\varphi = 0.40 \pm 0.05$).
2. All three MVT estimators return significant values — 28.2 ± 2.3 ms (canonical windowed), 353 ± 98 ms (Haar), 215 ± 17 ms (CWT) — and the CWT lands on the same grid rung for the fourth burst running; estimator-independence is falsified as stated, and CWT values are quoted only with the grid-quantization label.
3. The window-scanned lag is $\tau = 0.68^{+0.05}_{-0.06} \pm 0.13_{\text{win}}$ s; $\tau/T_{90} \approx 5.4\%$, between the two earlier measured bursts — the lag-width relation across the campaign remains non-constant.
4. Nine-block spectroscopy repeats the epoch pattern: the synchrotron line of death is violated in

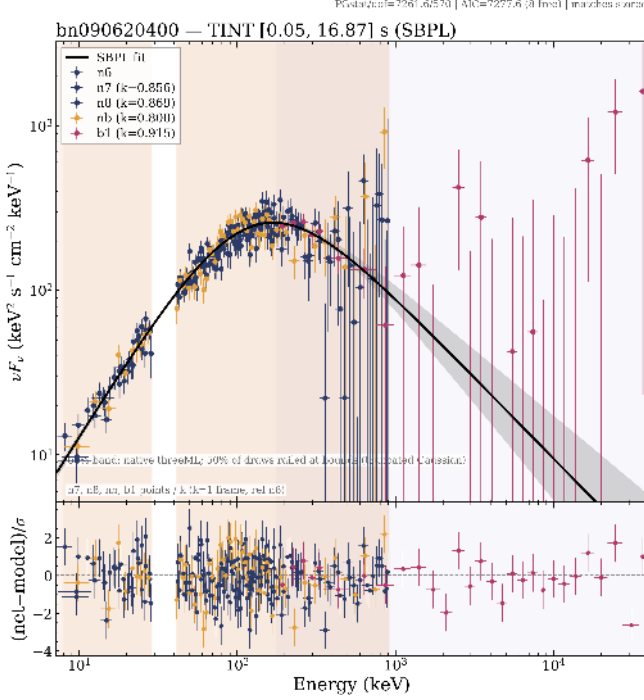


Figure 9. Integrated-spectrum winner: SBPL with a single break at 131 keV. 68% error band from propagated posterior draws; PGstat/dof and EACs stamped on-figure.

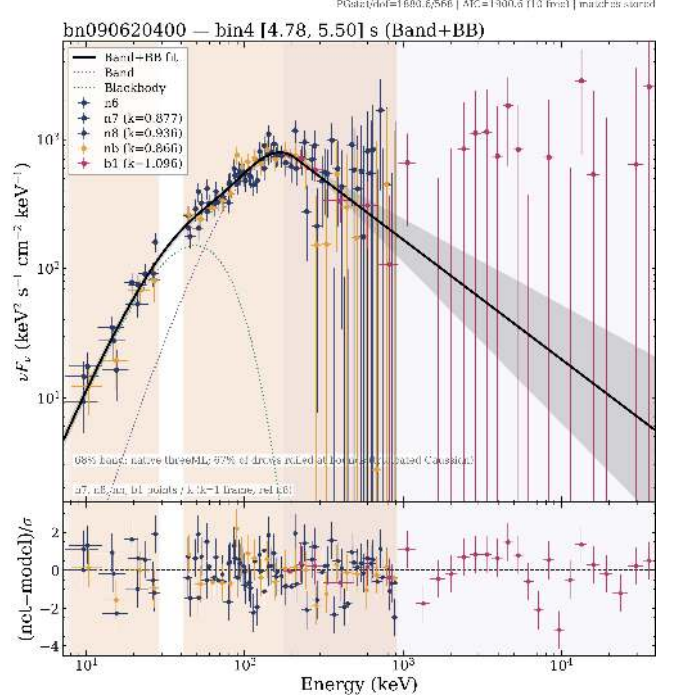


Figure 11. Block 4: Band+BB winner ($kT = 12.4$ keV, components dotted) — a $\Delta\text{AIC} = 0.4$ tie with plain Band, reported as a weak candidate only.

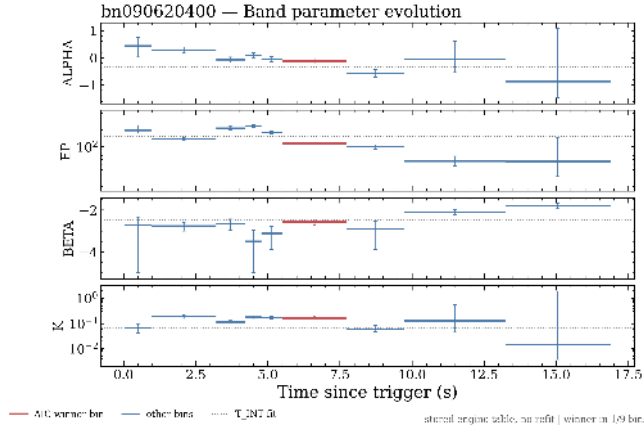


Figure 10. Band-family parameter evolution across the nine blocks: the α line-of-death crossing and the non-monotonic E_p track.

the rise/peak blocks (α up to +0.43) and respected in the decay (-0.87 by the end), with E_p tracking the light curve non-monotonically ($196 \rightarrow 237 \rightarrow 56$ keV).

5. The integrated spectrum is a one-break SBPL ($E_b = 131^{+18}_{-15}$ keV), tied with Band; the two-break DSBPL does not contend, in contrast to burst #4.
6. Three BB-bearing block winners survive as: one weak candidate ($kT = 12.4$ keV, tie), one *edge*

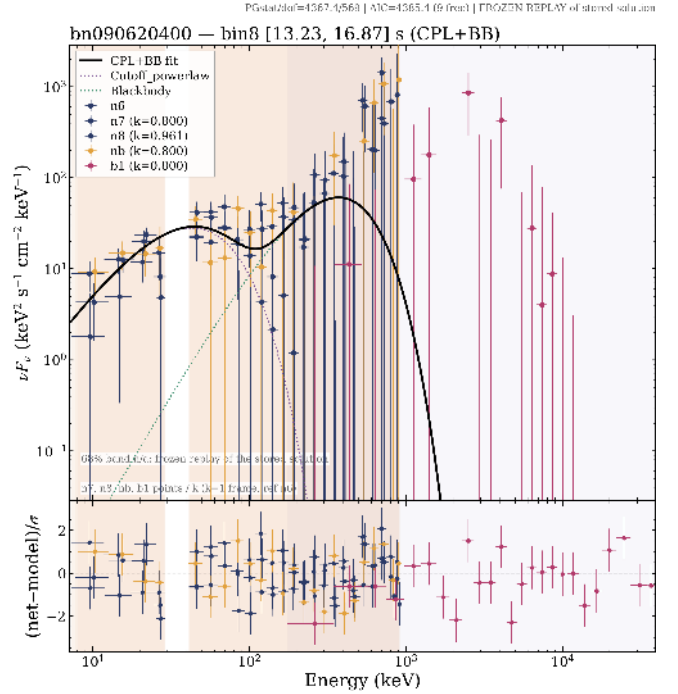


Figure 12. Block 8: CPL+BB tie ($kT = 95 \pm 33$ keV) — the uninformative high-side counterpart.

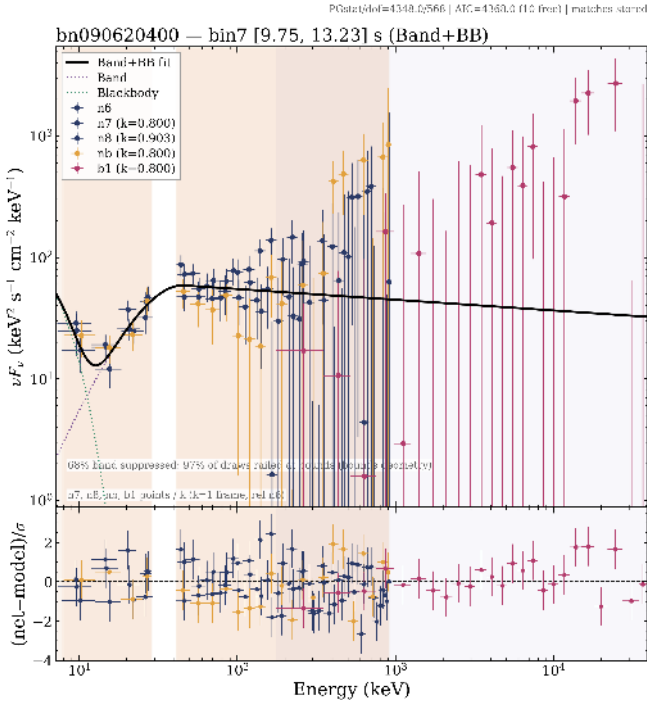


Figure 13. Block 7: the $kT = 1.10$ keV blackbody whose Wien peak (4.3 keV) falls below the 8 keV fit boundary — the edge-artifact signature; not a detection.

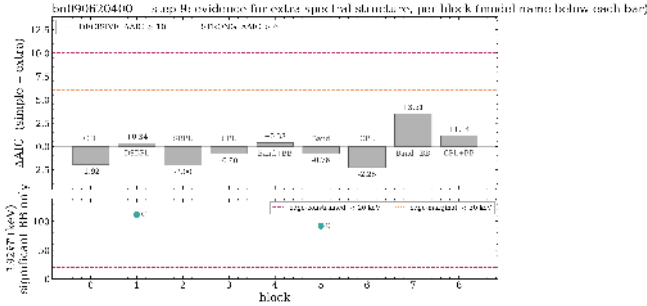


Figure 14. Step-9 scorecard: per-block evidence for extra spectral structure (top) and the $3.92 kT$ Wien-peak edge check (bottom; markers require a nested LRT ≥ 9.2 and appear only in blocks 1 and 5, both edge-clean — block 7’s $kT = 1.10$ keV blackbody falls below that gate, and its edge diagnosis is made in the text from the fit table).

artifact ($kT = 1.10$ keV, Wien peak below the fit boundary — the campaign’s edge gate working as designed), and one uninformative high-side tie. Zero robust thermal candidates.

7. Standing caveats: n_b ’s effective-area factor rails at 0.80; all numbers are provisional pending the PI’s campaign review.

APPENDIX

A. ALL-MODEL MONTAGES

Complete 24-model montages for the integrated spectrum and all nine blocks (Figures 15–24); the winner panel is marked in each; per-bin parameter tables accompany the paper products.

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Figure 15. Integrated spectrum: all 24 models.

bn090620400 — bin0 [0.05, 0.96] s — 24 models, AIC-ordered — winner CPL

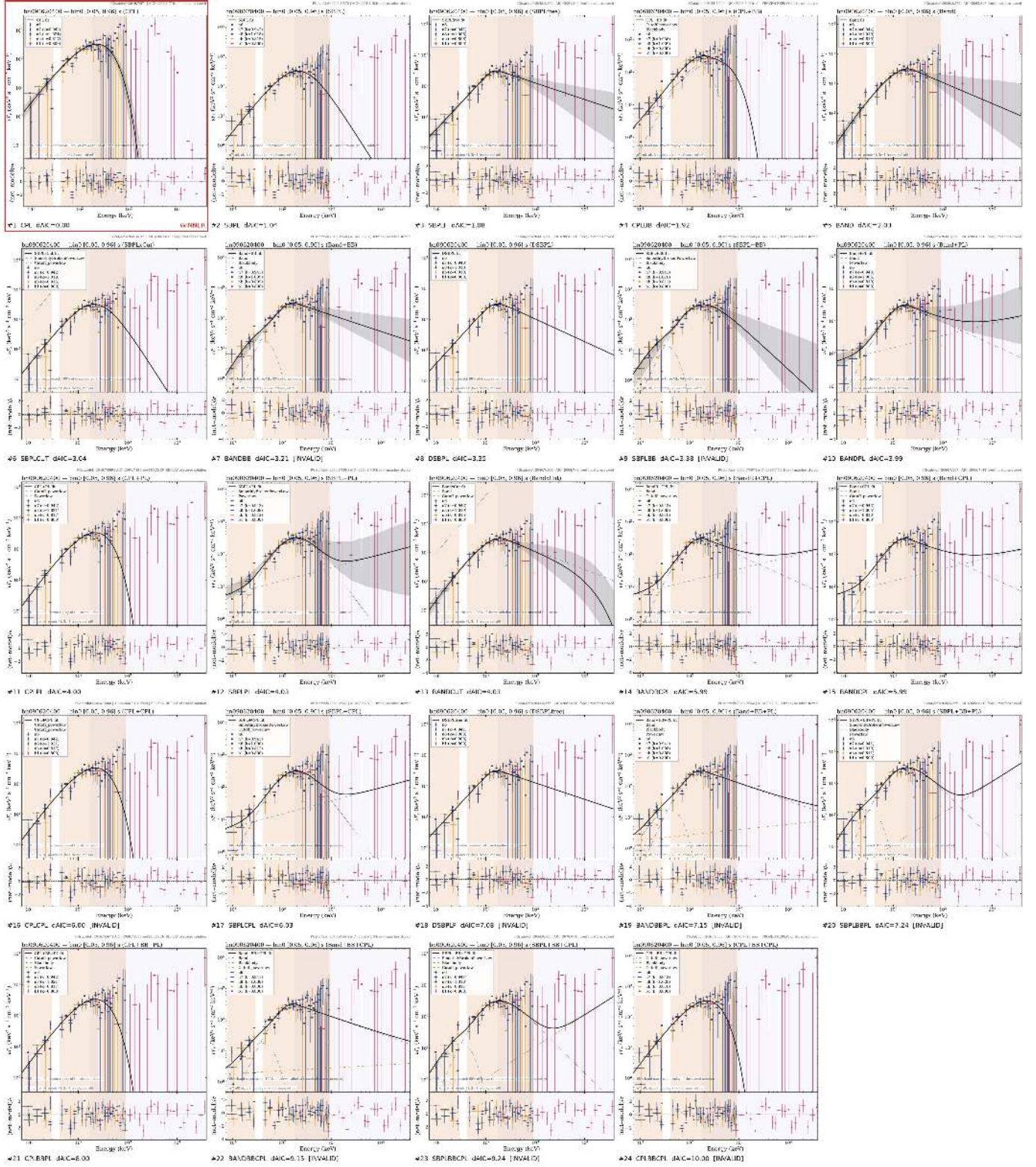


Figure 16. Block 0: all 24 models.

bn090620400 — bin1 [0.96, 3.21] s — 24 models, AIC-ordered — winner DSBPL

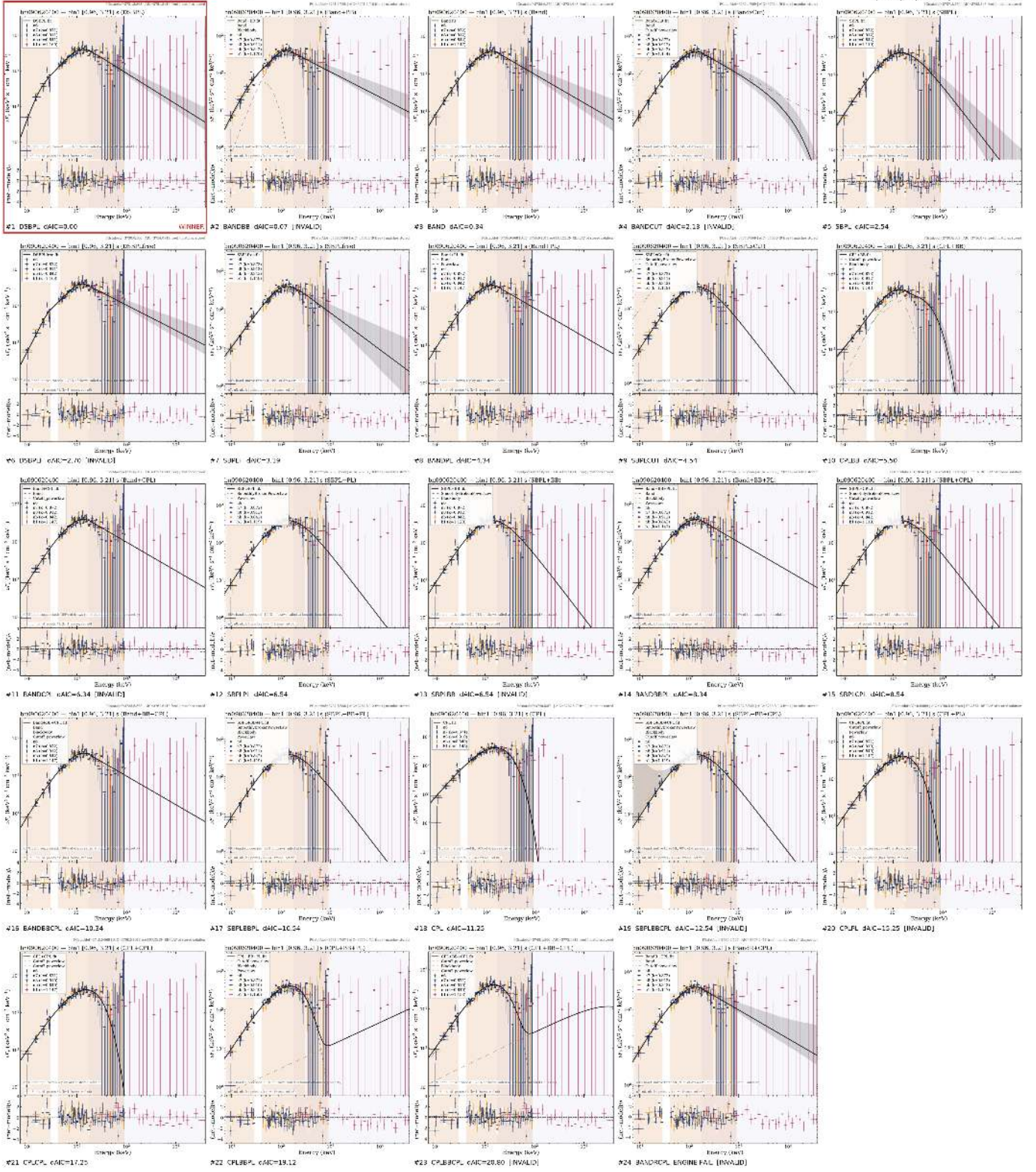


Figure 17. Block 1: all 24 models.

bn090620400 — bin2 [3.21, 4.21] s — 24 models, AIC-ordered — winner SBPL

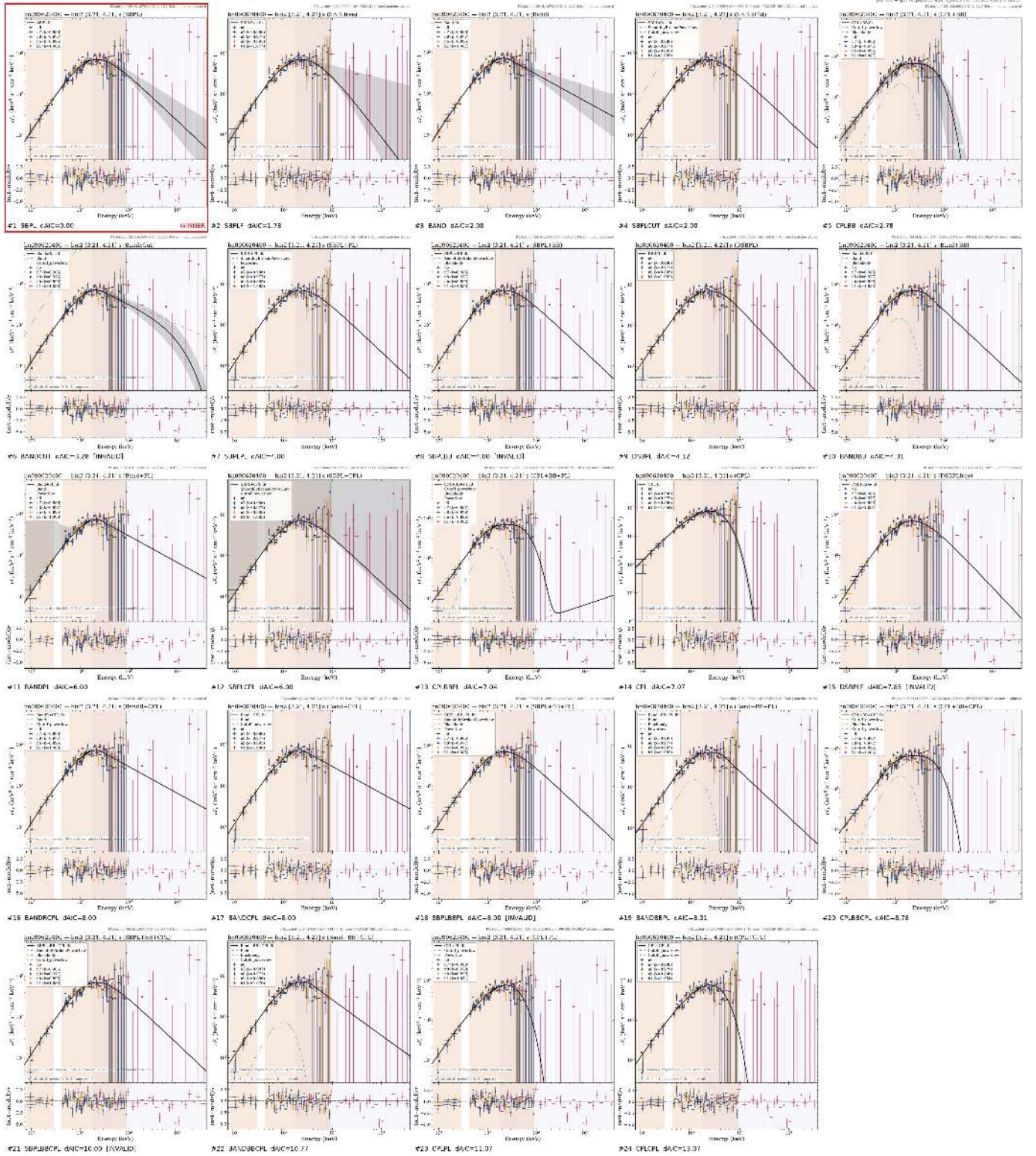


Figure 18. Block 2: all 24 models.

bn090620400 — bin3 [4.21, 4.78] s — 24 models, AIC-ordered — winner CPL

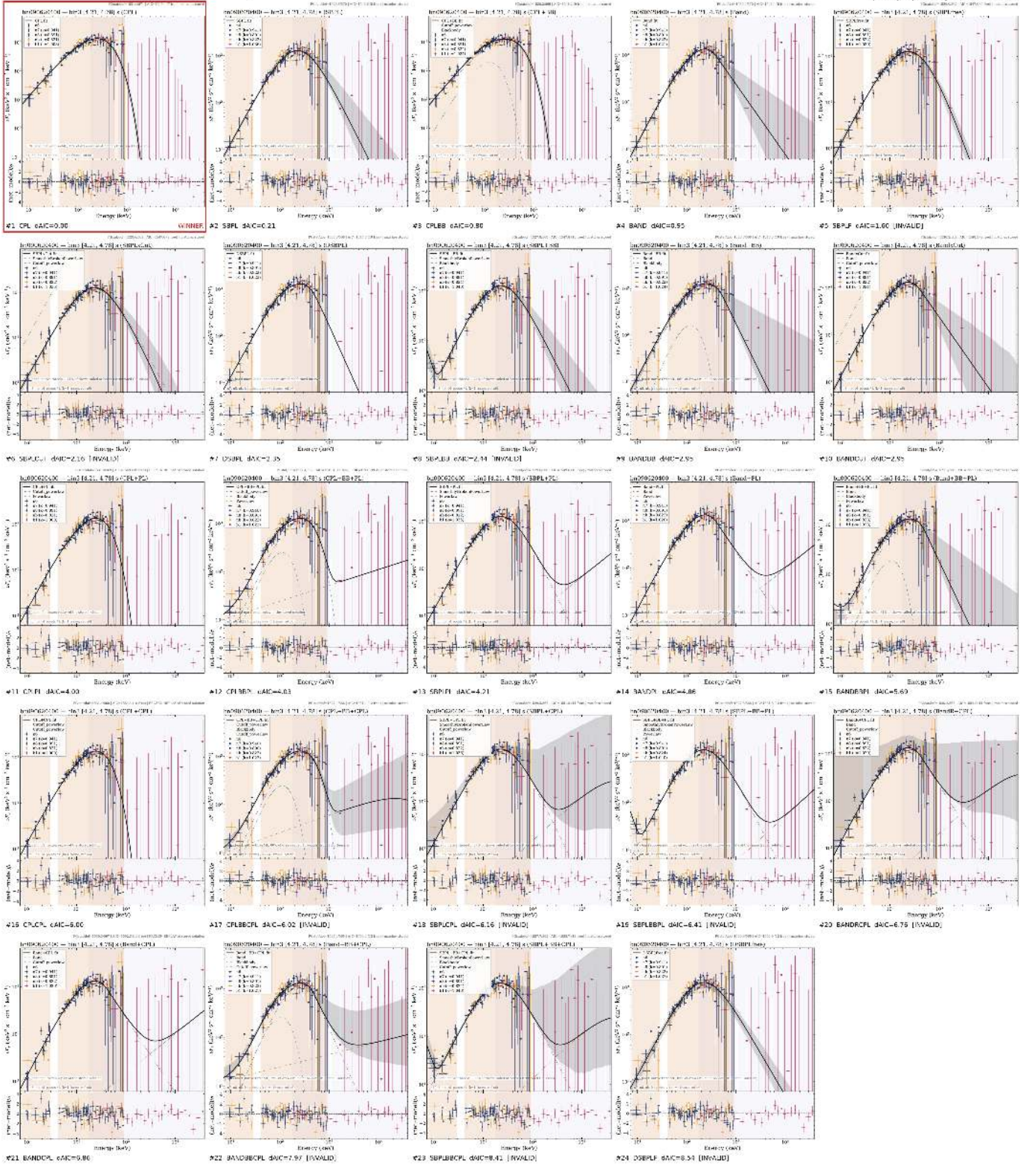


Figure 19. Block 3: all 24 models.

bn090620400 — bin4 [4.78, 5.50] s — 24 models, AIC-ordered — winner BANDBB

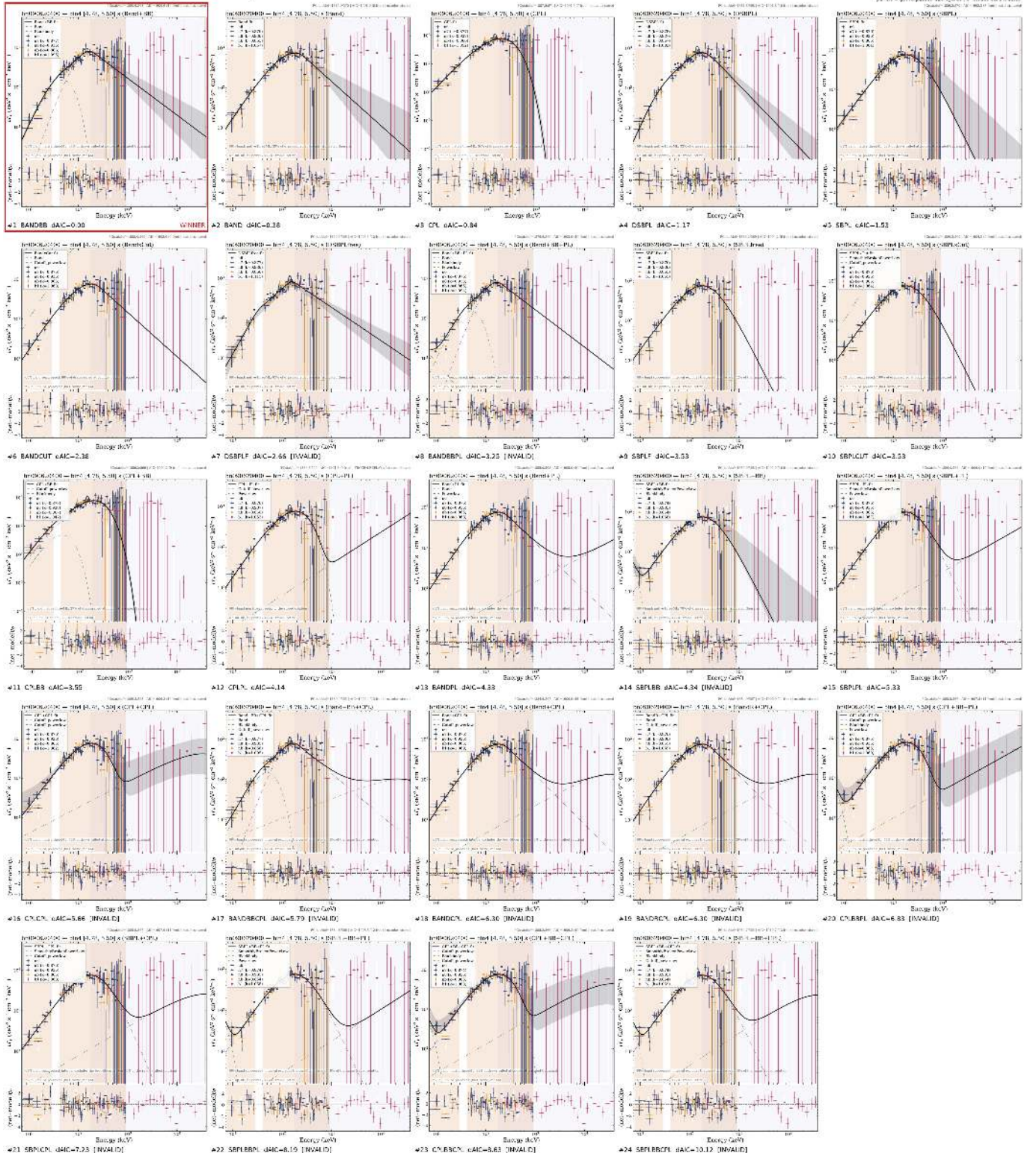


Figure 20. Block 4: all 24 models.

bn090620400 — bin5 [5.50, 7.74] s — 24 models, AIC-ordered — winner BAND



Figure 21. Block 5: all 24 models.

bn090620400 — bin6 [7.74, 9.75] s — 24 models, AIC-ordered — winner CPL

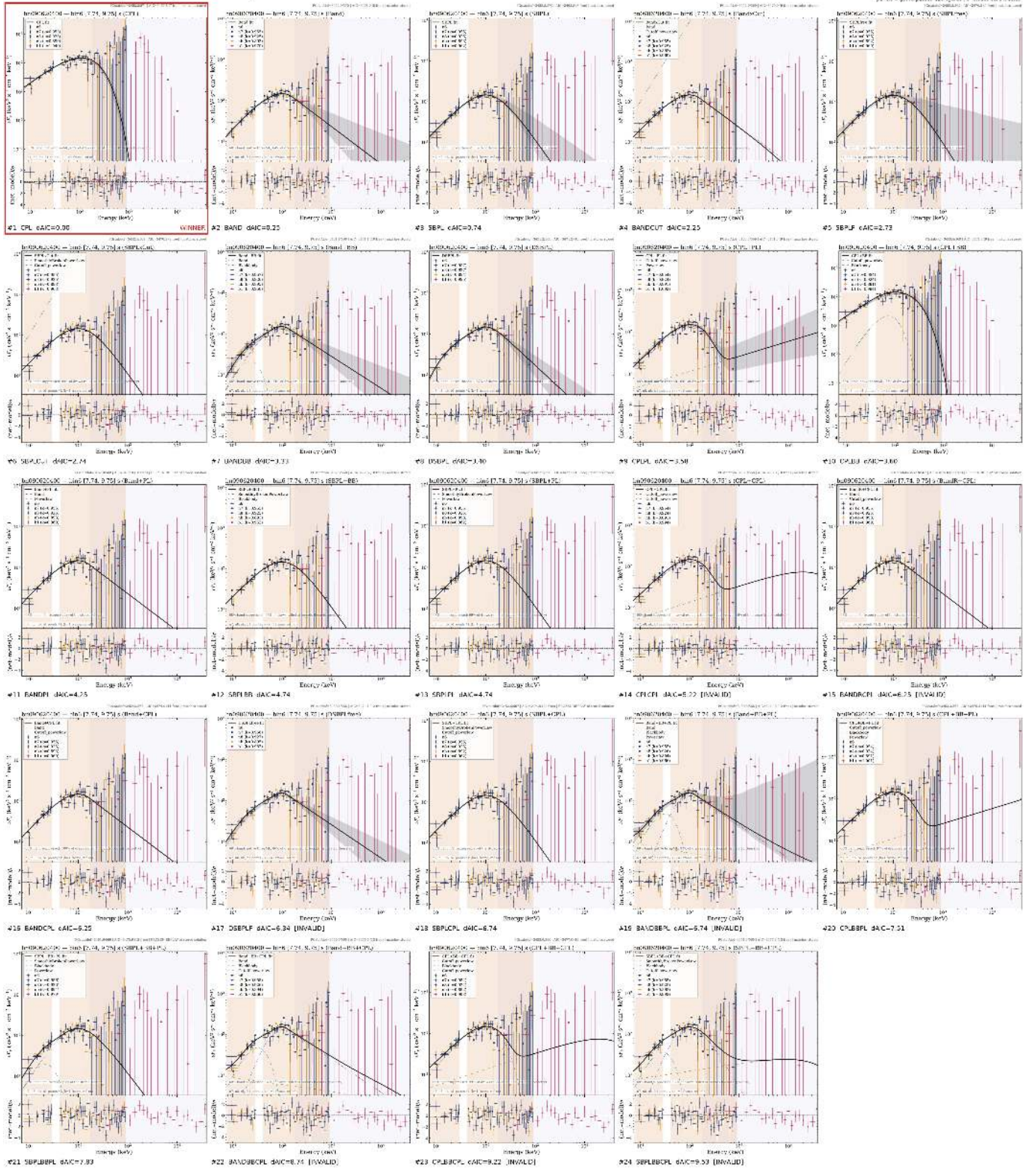


Figure 22. Block 6: all 24 models.

bn090620400 — bin7 [9.75, 13.23] s — 24 models, AIC-ordered — winner BANDBB

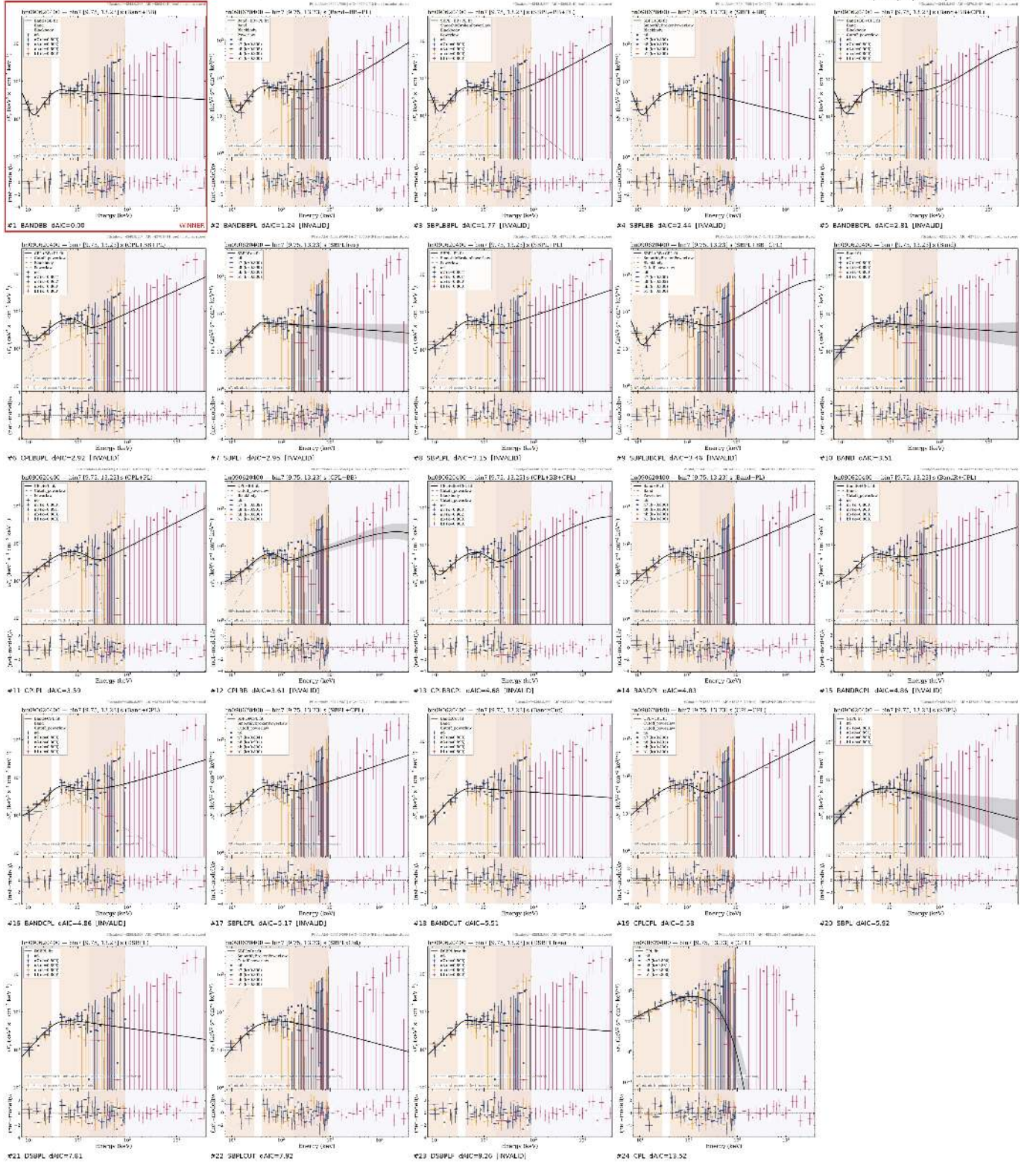


Figure 23. Block 7: all 24 models.

bn090620400 — bin8 [13.23, 16.87] s — 24 models, AIC-ordered — winner CPLBB



Figure 24. Block 8: all 24 models.

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